SI 315

Models of Social Information Theory

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Logistics

- No AI-generated submissions to assignments
- Required attendance at both lectures and discussions
- Two exams, final is non-cumulative
- iClicker both participation and correctness
- Lowest two grades dropped from each of assignments, quizzes, discussion attendance, lecture attendance

Contents

1 Introduction to Networks **2**

1.1 Strong and Weak Ties . . . 2
1.2 Bridges and Local Bridges . . . 2
1.3 Clustering . . . 2
1.4 Strong Triadic Closure Property . . . 3
1.5 Homophily . . . 3
1.6 Affiliation Networks . . . 3
1.7 Social Affiliation Networks . . . 3
1.7.1 Closure in Social Affiliation Networks . . . 4
1.8 Signed Networks . . . 4

2 Balance **4**

2.1 Balance Theory . . . 4
2.2 Weak Balance Theorem . . . 4

3 Game Theory **5**

3.1 Strategic-Form Games . . . 5
3.2 Dominant Strategies . . . 5
3.3 Example Games . . . 5
3.3.1 Presentation vs. Exam . . . 5
3.3.2 Prisoner's Dilemma . . . 5
3.3.3 A New Product . . . 5
3.3.4 Three Clients . . . 6
3.4 Nash Equilibrium . . . 6
3.5 Focal Points . . . 6
3.6 Mixed Strategies . . . 6
3.6.1 Finding a Mixed-Strategy Equilibrium . . . 6
3.6.2 Solving for an Unknown Payoff . . . 7

3.6.3 Probabilities from a Complete Payoff Matrix . . . 7
3.7 Matching Pennies . . . 8
3.8 Coordination Game . . . 8
3.9 Network Traffic . . . 8
3.10 Auctions . . . 9
3.11 Bipartite Matching . . . 9
3.11.1 Weighted Edges . . . 9

1 Introduction to Networks

1.1 Strong and Weak Ties

- Strong ties: frequent interaction, trust, emotional closeness
- Weak ties: infrequent interaction, limited intimacy
- Strong ties: overlapping social circles
- Weak ties: bridges between disconnected groups
- Weak ties: novel information and opportunities

1.2 Bridges and Local Bridges

- Bridge: an edge whose removal disconnects the network into separate components
- Local bridge: an edge whose endpoints do not have any direct ties to the same neighboring nodes
- Local bridge: removing it increases the distance between its endpoints, giving it a span of at least 3
- Span of a bridge: the distance between the endpoints after the edge is removed
- Bridges and local bridges: often connect otherwise distant or disconnected regions of the network

1.3 Clustering

- Clustering: connected neighbors also connected to one another
- Local clustering coefficient: connected neighbor pairs divided by possible neighbor pairs
- Formula: $C_i = \frac{2e_i}{k_i(k_i-1)}$
- e_i: edges among node i's neighbors
- Network clustering coefficient: average local coefficient

1.4 Strong Triadic Closure Property

- Two strong ties sharing a node: likely third edge
- Strong ties around a node: mutually connected neighbors
- Open triad: two strong ties without a connecting edge
- Open triads: pressure toward closure
- Closed triads: higher clustering coefficients

1.5 Homophily

- Homophily: the tendency for people to form ties with similar others
- Similarity may involve age, gender, education, location, interests, or beliefs
- Homophily can produce clustered groups with shared characteristics
- Network ties can reinforce existing similarities through social influence
- Observed similarity may result from selection, influence, or both
- Let p and q be the proportions of two groups, with $p + q = 1$
- Under random mixing, the probability of a cross-group tie is $2pq$
- Cross-group tie probability c under random mixing: expected to be $2pq$
- Homophily where the observed cross-group tie probability is less than expected: $c < 2pq$

1.6 Affiliation Networks

- Affiliation network: a network connecting people to the groups, events, organizations, or activities they participate in
- Two-mode network: contains two types of nodes, such as individuals and affiliations
- One-mode projection: connects two people when they share an affiliation
- Shared affiliations can create indirect social ties and opportunities for interaction
- Affiliations can reveal overlapping communities and patterns of participation

1.7 Social Affiliation Networks

- Social affiliation network: represents relationships between people and the groups, events, or organizations with which they are affiliated
- Membership links connect each person to the affiliations they share
- Shared affiliations can create indirect opportunities for social interaction and information flow
- One-mode projection: connects people who share at least one affiliation
- Projection ties may be weighted by the number of affiliations two people share
- Affiliation networks can reveal overlapping communities, central participants, and patterns of group involvement

1.7.1 Closure in Social Affiliation Networks

- Triadic: nodes with a common neighbor tend to form a direct connection, closing the triad
- Group/focal: nodes connected to the same group or affiliation tend to form direct connections, closing the group-based triad
- Membership: nodes are more likely to join groups that their neighbors are already members of

1.8 Signed Networks

- Signed network: a network in which edges have positive or negative signs, representing friendly or antagonistic relationships
- Positive tie: indicates friendship, trust, or alliance
- Negative tie: indicates enmity, distrust, or conflict
- Balance theory: suggests that triads tend to evolve toward a balanced state, where the product of the signs of the edges is positive
- Balanced triads: all positive edges or two negative edges and one positive edge
- Unbalanced triads: one negative edge and two positive edges or all negative edges

2 Balance

2.1 Balance Theory

- Balance theory: suggests that triads tend to evolve toward a balanced state, where the product of the signs of the edges is positive
- Balanced triads: 3 positive edges or 1 positive edge
- Unbalanced triads: one negative edge and two positive edges or all negative edges
- Balanced network: all triads are balanced

2.2 Weak Balance Theorem

- Weak balance theorem: in a signed network, if all triads are weakly balanced (no triad has exactly two negative edges), the network can be partitioned into two mutually antagonistic groups with positive ties within each group and negative ties between groups
- Weakly balanced triads: triads that have 3 positive edges, 1 positive edge, or 0 positive edges
- Weakly balanced network: all triads in the network are weakly balanced

3 Game Theory

3.1 Strategic-Form Games

- Game: players, available strategies, and payoffs for each strategy profile
- Strategy profile: one strategy chosen by each player
- Payoff: value or outcome a player receives from a strategy profile
- Best response: strategy that maximizes a player's payoff given the other players' strategies
- Strict best response: strategy that gives a strictly higher payoff than every alternative, given the other players' strategies
- More than one best response possible when strategies tie for the highest payoff

3.2 Dominant Strategies

- Dominant strategy: best response to every possible strategy profile of the other players
- Strictly dominant strategy: gives a strictly higher payoff than every alternative for every possible strategy profile of the other players
- Strictly dominant strategy: rational choice regardless of what the other players do
- A game may have no dominant strategy, or may have dominant strategies for some or all players

3.3 Example Games

3.3.1 Presentation vs. Exam

- Compare each player's available choices and payoffs in every outcome
- Mark each player's best response to each choice of the other player
- Use the payoff matrix to determine whether choices align or conflict

3.3.2 Prisoner's Dilemma

- Two players independently choose to cooperate or defect
- Defecting: strictly dominant strategy for each player
- (Defect, Defect): Nash equilibrium, despite mutual cooperation giving both players a better outcome
- Individual incentives can produce an outcome worse for both players than cooperation

3.3.3 A New Product

- Analyze the available choices and payoffs for each player using the lecture payoff matrix
- Check how one player's best response changes with the other player's choice
- Use best responses to identify any Nash equilibria

3.3.4 Three Clients

- Identify the players, available strategies, and payoffs from the lecture example
- For each player, compare payoffs across their choices while holding the other players' choices fixed
- Mark mutual best responses to locate Nash equilibria

3.4 Nash Equilibrium

- Two-player strategy profile (S, T): Nash equilibrium if S is a best response to T and T is a best response to S
- General case: every player's strategy is a best response to the strategies chosen by all other players
- At equilibrium: no player can improve their payoff by changing strategy alone
- Finite-game existence theorem: every game with finitely many players and finitely many strategies has at least one Nash equilibrium in mixed strategies
- Pure-strategy Nash equilibrium: equilibrium using a single strategy per player; not guaranteed to exist

3.5 Focal Points

- Focal point: salient outcome that helps players coordinate without communication
- Salience may come from conventions, shared expectations, or a distinctive feature of the game
- Focal points can help players select among multiple coordination-game equilibria

3.6 Mixed Strategies

- Mixed strategy: probability distribution over a player's available pure strategies
- Pure strategy: one action chosen with probability 1
- Expected payoff: probability-weighted average payoff across possible outcomes
- For strategy profile (s_i, s_{-i}) drawn from mixed strategies:

$$\mathbb{E}[u_i] = \sum_{s_i, s_{-i}} \Pr(s_i) \Pr(s_{-i}) u_i(s_i, s_{-i})$$

- Mixed-strategy Nash equilibrium: each player's probability distribution is a best response to the others' distributions
- In equilibrium, every pure strategy used with positive probability gives the player the same expected payoff

3.6.1 Finding a Mixed-Strategy Equilibrium

- Assign probabilities to one player's strategies
- Choose probabilities that make the other player indifferent among their pure strategies
- Repeat in the opposite direction to find the first player's equilibrium probabilities
- Indifference makes mixing optimal; a player would otherwise put all probability on a higher-payoff pure strategy

3.6.2 Solving for an Unknown Payoff

- **Indifference principle**: in a mixed-strategy NE, each player's mix makes the *other* player indifferent among strategies played with positive probability
- Steps:
  1. Get equilibrium probabilities (usually given)
  2. Write indifference condition for the player whose payoffs are known: uses the opponent's mix
  3. Write indifference condition for the player with the unknown payoff x: $\mathbb{E}[u_1(U)] = \mathbb{E}[u_1(D)]$
  4. Solve for x: linear, one equation, one unknown
- Unknown payoff of player i appears in i's indifference condition, which uses the *opponent's* probabilities
- If x also affects the opponent's mix: use both indifference equations with given probabilities
- If no probabilities given: treat p, q as unknowns; two indifference equations, solve for p, q, x
- Check: all probabilities in $[0, 1]$; no profitable deviation to a pure strategy outside the mix

Example

- Payoffs (Player 1, Player 2):

	L	R
U	$x, 2$	$0, 1$
D	$1, 0$	$3, 3$

- Given: Player 2 plays L with probability $q = \frac{3}{4}$
- Player 1 indifferent between U and D:

$$\mathbb{E}[u_1(U)] = qx + (1-q)\cdot 0 = \tfrac{3}{4}x$$
$$\mathbb{E}[u_1(D)] = q\cdot 1 + (1-q)\cdot 3 = \tfrac{3}{4} + \tfrac{3}{4} = \tfrac{3}{2}$$

- Set equal: $\frac{3}{4}x = \frac{3}{2}$, so $\boxed{x = 2}$

3.6.3 Probabilities from a Complete Payoff Matrix

- p = Pr(Player 1 plays U); q = Pr(Player 2 plays L)
- p comes from **Player 2's** indifference; q comes from **Player 1's** indifference
- Own payoffs only determine what the opponent does; given q is not needed to find p
- Check dominated strategies first: strictly dominated gets probability 0; fully mixed NE needs no dominant strategy
- Valid only if $0 < p, q < 1$

Example

- Payoffs (Player 1, Player 2):

	L	R
U	$4, 3$	$1, 1$
D	$2, 0$	$5, 4$

- Find p: Player 2 indifferent between L and R

$$\mathbb{E}[u_2(L)] = 3p + 0(1-p) = 3p$$
$$\mathbb{E}[u_2(R)] = 1p + 4(1-p) = 4 - 3p$$

- $3p = 4 - 3p$, so $\boxed{p = \frac{2}{3}}$
- Find q: Player 1 indifferent between U and D

$$\mathbb{E}[u_1(U)] = 4q + 1(1-q) = 1 + 3q$$
$$\mathbb{E}[u_1(D)] = 2q + 5(1-q) = 5 - 3q$$

- $1 + 3q = 5 - 3q$, so $\boxed{q = \frac{2}{3}}$
- Mixed NE: $p = q = \frac{2}{3}$
- Sanity check: $q = \frac{1}{2}$ is not an equilibrium value; $\mathbb{E}[u_1(U)] = 2.5 < 3.5 = \mathbb{E}[u_1(D)]$, so Player 1 strictly prefers D

3.7 Matching Pennies

- Each player chooses Heads or Tails
- Player 1 wins when choices match; Player 2 wins when choices differ
- No pure-strategy Nash equilibrium: either player can switch and improve their payoff
- Let q be the probability Player 2 chooses Heads
- Player 1's expected payoff from Heads: $q - (1-q) = 2q - 1$
- Player 1's expected payoff from Tails: $(1-q) - q = 1 - 2q$
- Player 1 is indifferent when $2q - 1 = 1 - 2q$, so $q = \frac{1}{2}$
- By symmetry, Player 1 also chooses Heads with probability $\frac{1}{2}$
- Mixed-strategy Nash equilibrium: each player chooses Heads and Tails with probability $\frac{1}{2}$

3.8 Coordination Game

- Players benefit from choosing strategies that match or coordinate
- Coordination games often have multiple Nash equilibria
- Equilibrium selection: players need expectations, conventions, or focal points to coordinate on one equilibrium
- A coordinated outcome may be better for everyone than a mismatched outcome

3.9 Network Traffic

- Players choose routes in a network to minimize their own travel time
- Travel time on a route depends on the number of players using it
- Nash equilibrium occurs when no player can reduce their travel time by unilaterally changing routes
- Equilibria may be inefficient compared to a centrally coordinated solution (Braess's paradox)

3.10 Auctions

- Types of auctions: ascending (English), descending (Dutch), first-price sealed-bid, second-price sealed-bid (Vickrey)
- Second-price sealed-bid auction: buyers are players, bids are strategies—assume each buyer has a true value v_i and submits a bid b_i; highest bidder "payoff" is $v_i - b_j$ (the money saved compared to their true value), where b_j is the second-highest bid; dominant strategy is to bid their true value $b_i = v_i$; truthful bidding leads to efficient allocation of the item.

3.11 Bipartite Matching

- Bipartite graph: graph whose vertices can be divided into two disjoint sets where no two vertices within the same set are adjacent
- Bipartite matching: set of edges in a bipartite graph such that no two edges share a common vertex
- Maximum bipartite matching: bipartite matching that contains the largest possible number of edges
- Constricted set: subset of vertices in one partition of a bipartite graph such that the number of neighbors in the other partition is strictly smaller than the number of vertices in the subset

3.11.1 Weighted Edges

- Weighted bipartite graph: bipartite graph where each edge has an associated weight
- Maximum weight bipartite matching (optimal assignment): bipartite matching that maximizes the sum of the weights of the selected edges